

NexaMart : AI Customer Support Agent Deployment

Self-Directed Business Analysis Case Study | 2026

Analysis Technique: SWOT Analysis

Prepared by: Safi Ullah Khan

Project Stage: Discovery / Initial Assessment

Document Version: 1.0

Date: September 2026

Status: Portfolio / Simulated Enterprise Case Study

Case Study Classification

This is a self-directed, simulated business analysis case study developed for professional learning and portfolio demonstration. NexaMart is a fictional organization created for this exercise. The analysis does not represent work performed for an actual client or employer.

Table of Contents

Section 1 : Executive Summary	2
Section 2 : Business Context	2
2.1 NexaMart	2
2.2 Current Situation	2
2.3 Proposed Initiative	2
Section 3 : Business Problem	3
3.1 Business problem	3
3.2 Business opportunity	3
3.3 Important assumption	3
Section 4 : Business Objective	3
4.1 Primary objective:	3
Section 5 : SWOT Definition	4
5.1 What is SWOT Analysis?	4
Section 6 : Analysis Scope	4
Section 7 : SWOT Interpretation	4
7.1 Key Strengths	5
7.2 Key Weaknesses	5
7.3 Key Opportunities	5
7.4 Key Threats	5
Section 8 : Strategic Implications	5
8.1 Strength → Opportunity	5
8.2 Weakness → Opportunity	6
8.3 Strength → Threat	6
8.4 Weakness → Threat	6
Section 9 : BA Recommendation	6
9.1 Recommendation 1 : Conduct knowledge assessment	6
9.2 Recommendation 2 : Conduct process analysis	6
9.3 Recommendation 3 : Identify automation candidates	6
9.4 Recommendation 4 : Define human escalation	7
9.5 Recommendation 5 : Assess integration readiness	7
Section 10 : Questions for Stakeholders	7
10.1 Product Owner	7
10.2 Customer Support Manager	7
10.3 IT	7
10.4 Security	7
10.5 Compliance/Legal	7
10.6 Operations	7
Section 11 : Assumptions & Dependencies	8
Section 12 : Conclusion	8

Section 1 : Executive Summary

Executive Summary

This case study evaluates the strategic suitability of deploying an AI-powered customer support agent at NexaMart, a fictional multinational e-commerce organization.

The proposed solution is intended to automate high-volume, repetitive customer-support interactions while maintaining human escalation for complex, sensitive or low-confidence cases.

A SWOT analysis was conducted to assess internal organizational capabilities and limitations alongside external opportunities and threats associated with the initiative.

The analysis identifies areas that may support the initiative, as well as potential barriers requiring further investigation before implementation. The findings are intended to inform subsequent discovery activities, including process analysis, requirements elicitation, risk assessment and feasibility evaluation.

Section 2 : Business Context

2.1 NexaMart

A fictional multinational e-commerce organization serving customers through digital channels and customer-support operations.

2.2 Current Situation

NexaMart currently handles customer inquiries through human-operated support channels. A significant proportion of interactions relate to repetitive requests such as order status, delivery information, return policies, refund status and account-related questions.

Customer-support representatives must frequently search across knowledge repositories and operational systems to resolve customer requests. This creates opportunities for process improvement and selective automation.

2.3 Proposed Initiative

NexaMart is exploring the deployment of an AI-powered customer-support agent capable of handling selected customer interactions, retrieving information from approved enterprise sources and escalating complex or low-confidence cases to human support representatives.

Section 3 : Business Problem

3.1 Business problem

NexaMart's customer-support operation may be experiencing avoidable workload and response-time pressure due to repetitive customer inquiries, manual information retrieval and multiple system interactions.

3.2 Business opportunity

Selective AI-assisted automation may allow NexaMart to handle appropriate low-complexity interactions more efficiently while enabling human representatives to focus on complex customer issues.

3.3 Important assumption

The actual volume, cost, response-time distribution, customer satisfaction impact and automation suitability would need to be validated during subsequent discovery.

Section 4 : Business Objective

4.1 Primary objective:

Assess whether NexaMart is organizationally and strategically positioned to pursue AI-enabled customer-support automation.

Potential objectives:

- Reduce repetitive support workload.
- Improve response times.
- Provide support outside traditional operating hours.
- Improve consistency of responses.
- Enable human agents to focus on complex cases.
- Maintain appropriate security, governance and escalation controls.

Section 5 : SWOT Definition

5.1 What is SWOT Analysis?

SWOT Analysis is a strategic analysis technique used to evaluate an initiative, organization, product or decision by examining internal Strengths and Weaknesses and external Opportunities and Threats.

Section 6 : Analysis Scope

Dimension	Scope
Organization	NexaMart
Initiative	AI Customer Support Agent
Analysis stage	Initial discovery
Perspective	Business + operational + technology
Stakeholders	Product, Customer Support, IT, Security, Compliance, Operations
Time context	2026
Objective	Assess strategic readiness and areas requiring further investigation

Section 7 : SWOT Interpretation

Strengths	Weaknesses
Existing CRM capability	Fragmented knowledge sources
Historical customer interaction data	Legacy system integrations
Existing customer-support workforce	Manual information retrieval
Established escalation processes	Limited AI governance capability
Opportunities	Threats
24/7 customer support	Incorrect AI responses
Reduced repetitive workload	Data privacy/security risks
AI-assisted human agents	Regulatory requirements
Multilingual support	Customer resistance
Faster response times	Reputational damage

7.1 Key Strengths

NexaMart's existing CRM infrastructure and established customer-support operation may provide a foundation for integrating AI capabilities without creating an entirely new support ecosystem.

7.2 Key Weaknesses

Fragmented knowledge sources and legacy integrations could limit AI effectiveness and increase implementation complexity.

7.3 Key Opportunities

AI-assisted handling of repetitive, low-complexity inquiries could potentially improve availability and allow human representatives to concentrate on higher-value interactions.

7.4 Key Threats

AI-generated inaccuracies, privacy/security concerns and inappropriate automation of sensitive cases could create operational and reputational risk.

Section 8 : Strategic Implications

8.1 Strength → Opportunity

Existing CRM infrastructure could be leveraged to integrate AI-assisted customer support into existing workflows.

8.2 Weakness → Opportunity

Knowledge fragmentation should be addressed before large-scale AI deployment because response quality depends on reliable and approved information sources.

8.3 Strength → Threat

Existing human support capabilities can be used as an escalation mechanism for AI uncertainty or sensitive customer interactions.

8.4 Weakness → Threat

Legacy integrations combined with limited AI governance could increase implementation and operational risk.

Section 9 : BA Recommendation

9.1 Recommendation 1 : Conduct knowledge assessment

Before AI deployment, assess:

- Knowledge sources
- Accuracy
- Ownership
- Duplication
- Currency
- Access permissions

9.2 Recommendation 2 : Conduct process analysis

Map the existing customer-support process.

9.3 Recommendation 3 : Identify automation candidates

Separate:

Low-risk repetitive queries

from

high-risk/complex queries.

9.4 Recommendation 4 : Define human escalation

Establish conditions under which the AI must transfer the interaction.

9.5 Recommendation 5 : Assess integration readiness

Review CRM, order-management, knowledge and authentication systems.

Section 10 : Questions for Stakeholders

10.1 Product Owner

- What business outcome are we expecting from AI support?
- Which customer journeys are currently the biggest pain points?

10.2 Customer Support Manager

- What proportion of cases are repetitive?
- Which cases require human judgment?

10.3 IT

- Which systems contain the information required by the AI?
- What integration mechanisms are currently available?

10.4 Security

- What customer information would the AI have access to?

10.5 Compliance/Legal

- Which interactions should never be automated?

10.6 Operations

- What happens when the AI cannot resolve an interaction?

Section 11 : Assumptions & Dependencies

ID	Assumption / Dependency	Impact	Validation Method
A01	CRM can support required integration	High	Technical assessment
A02	Knowledge content is sufficiently reliable	High	Knowledge audit
A03	AI can be restricted to approved information	High	Architecture/security review
A04	Customers will accept AI support	Medium	Customer research
A05	Human escalation process exists	High	Process workshop

Section 12 : Conclusion

The analysis indicates that NexaMart may have a viable foundation for AI-enabled customer-support automation due to its existing customer-support infrastructure and potential opportunities for handling repetitive interactions.

However, the initiative presents material considerations around knowledge quality, legacy integrations, AI governance, data security and human escalation.

Based on the current analysis, a controlled discovery and feasibility phase is recommended before committing to full-scale deployment.

Section 13 : Next Steps

The next recommended BA activity is to model the current customer-support process, identify automation opportunities and establish requirements for a controlled AI-supported operating model.